

	Catalan numbers	Little Schröder numbers
	(binary branching)	(super-Catalan; OEIS A001003)
Counts	rooted ordered trees with $k+1$ leaves, every internal node exactly 2 children	rooted ordered trees with n leaves, every internal node ≥ 2 children
Values ($k, n=0, 1, 2, \dots$)	1, 1, 2, 5, 14, 42, 132, $\dots$	1, 1, 3, 11, 45, 197, 903, $\dots$
Closed form	$C_k = \frac{1}{k+1} \binom{2k}{k}$	(no comparably simple closed form)
Recurrence	$C_0 = 1, \quad C_{k+1} = \sum_{i=0}^k C_i C_{k-i}$	$s_1 = s_2 = 1,$ $(n+1) s_{n+1} = 3(2n-1) s_n - (n-2) s_{n-1}$ for $n \geq 2$
Generating function	$\sum_{k \geq 0} C_k x^k = \frac{1 - \sqrt{1 - 4x}}{2x}$	$\sum_{n \geq 1} s_n x^n = \frac{1 + x - \sqrt{1 - 6x + x^2}}{4}$

Same indexing convention as `catalan.py`: the right-hand column's n counts leaves directly ($s_n = \text{all_trees_new}(n)$); the left-hand column's k is the standard Catalan index (trees with $k+1$ leaves), since that is how Catalan numbers are conventionally stated. Both recurrences and generating functions verified numerically against the known sequences before typesetting.